AGRIGUARD IoT: SOLAR-POWERED SMART SCARECROW SYSTEM

User Acceptance Testing (UAT) Survey

Based on ISO/IEC 25010 Software Quality Model

Consent Statement

Your participation in this survey is voluntary. All responses will be kept confidential and used solely for academic research purposes. By completing this survey, you agree to participate in this study. You may withdraw at any time without penalty.

Part I. Respondent Profile

Age: __________

Sex

☐ Male

☐ Female

☐ Prefer not to say

Occupation: ☐ Rice Farmer ☐ Farm Owner ☐ Agricultural Worker ☐ Other: __________

Highest Education: ☐ No Formal Education ☐ Elementary ☐ High School

☐ Vocational/Technical ☐ College

Years of Farming: ☐ Less than 5 years ☐ 5–10 years ☐ 11–20 years ☐ 21 years and above

Smartphone Usage: ☐ Daily ☐ Weekly ☐ Occasionally ☐ Rarely

Instructions

Please rate each statement based on your actual experience using the AgriGuard IoT system and its Android application.

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

N/A – Not Applicable / Feature Not Used

A. Functional Suitability

Checks if the system does what the user needs.

Statement	1	2	3	4	5	N/A
A1. The camera can detect birds as expected.						
A2. The system can identify the birds correctly.						
A3. The scarecrow responds when a bird is detected.						
A4. The sound used to scare birds works properly.						
A5. The app shows the bird detection information I need.						
A6. The system saves the bird type, date, and time of each detection.						
A7. I can see the saved bird detection records in the app.						
A8. The system helps me check bird activity in the farm.						
A9. The system is useful for helping protect rice crops from birds.						

B. Performance Efficiency

Checks if the system works fast and smoothly.

Statement	1	2	3	4	5	N/A
B1. The system detects birds quickly.						
B2. The app opens and responds quickly.						
B3. The system works smoothly during use.						
B4. The Raspberry Pi and camera work together with minimal delay						
B5. The system can still work when there is no internet.						
B6. The system uses power efficiently						

C. Usability

Checks if the system is easy to learn and use.

Statement	1	2	3	4	5	N/A
C1. The AgriGuard IoT system is easy to learn.						
C2. The app is easy to navigate and use.						

C3. The app is clear and easy to understand.						
C4. The buttons and icons are easy to understand.						
C5. I can easily find the features I need.						
C6. The information shown in the app is easy to understand.						
C7. I can use the system without special technical skills.						

D. Reliability

Checks if the system works properly every time and keeps saved records.

Statement	1	2	3	4	5	N/A
D1. The system works properly during use.						
D2. The system rarely stops, freezes, or closes by itself.						
D3. The bird detection feature works properly during use.						
D4. Saved bird records are still available when I need them.						
D5. The app shows my saved records correctly when I open it again.						
D6. The system still works properly without internet.						

E. Security

Checks if the user's information and saved data feel safe.

Statement	1	2	3	4	5	N/A
E1. The system keeps bird records on the local network.						
E2. The system works without internet.						
E3. The app does not publicly share bird records.						
E4. Bird records are stored only in the system.						
E5. The app allows users to manually delete incorrect detection records, such as false detections of people.						

F. Portability

Checks if the app can work well on different Android phones.

Statement	1	2	3	4	5	N/A
F1. The app works properly on my phone.						
F2. The app is easy to install.						
F3. I can use the app without help from someone else.						
F4. The app works properly on different Android phones.						
F5. The system can be used properly in the farm.						

Part III. Overall Assessment

Statement	1	2	3	4	5	N/A
O1. Overall, I am satisfied with AgriGuard IoT.						
O2. I would recommend AgriGuard IoT to other rice farmers.						
O3. AgriGuard IoT can help reduce damage caused by birds.						
O4. AgriGuard IoT can help save time when checking for birds.						
O5. AgriGuard IoT is useful compared with checking for birds by hand.						

Part IV. Open-Ended Questions

1. What feature did you like most? Why?

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2. Did you have any problems while using the system?

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3. What can we improve in the system?